Topology: MAT307/M405/MAT803 : - Problem Set 2

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1. Whether each of the following subsets of $\mathbb{R}^2$ can be written as a Cartesian product $A \times B$, where $A, B \subset \mathbb{R}$?
   (i) $\{(x, y) \mid x \text{ is an integer}\}$.
   (ii) $\{(x, y) \mid 0 \leq x < 1\}$.
   (iii) $\{(x, y) \mid x < y\}$.
   (iv) $\{(x, y) \mid x \text{ is an integer, and } y \text{ is not a rational number }\}$.
   (v) $\{(x, y) \mid x^2 + y^2 \leq 2026\}$.

2. Let $f : A \to B$ be a function, with $A_0 \subseteq A$ and $B_0 \subseteq B$.
   (i) Show that $A_0 \subseteq f^{-1}(f(A_0))$, and that equality holds for every subset $A_0 \subseteq A$ if and only if f is injective.
   (ii) Show that $f(f^{-1}(B_0)) \subseteq B_0$, and that equality holds for every subset $B_0 \subseteq B$ if and only if f is surjective.

3. Let $f : A \to B$ be a function, and $B_0, B_1 \subseteq B$. Show that preimages preserve inclusions, unions, intersections, and set differences by proving the following:
   (i) If $B_0 \subseteq B_1$, then $f^{-1}(B_0) \subseteq f^{-1}(B_1)$.
   (ii) $f^{-1}(B_0 \cup B_1) = f^{-1}(B_0) \cup f^{-1}(B_1)$.
   (iii) $f^{-1}(B_0 \cap B_1) = f^{-1}(B_0) \cap f^{-1}(B_1)$
   (iv) $f^{-1}(B_0 \setminus B_1) = f^{-1}(B_0) \setminus f^{-1}(B_1)$

4. Let $f : A \to B$ be a function, and let $A_0, A_1 \subseteq A$.
   (i) Show that images preserve inclusions and unions:
      (a) If $A_0 \subseteq A_1$, then $f(A_0) \subseteq f(A_1)$.
      (b) $f(A_0 \cup A_1) = f(A_0) \cup f(A_1)$.
   (ii) Show that images satisfy only one-way inclusions for intersections and set differences in general:
      (a) $f(A_0 \cap A_1) \subseteq f(A_0) \cap f(A_1)$.
      (b) $f(A_0) \setminus f(A_1) \subseteq f(A_0 \setminus A_1)$.
   (iii) Prove that equality holds in both (ii)(a) and (ii)(b) for all subsets $A_0, A_1 \subseteq A$ if and only if f is injective.

5. Let $\sim$ be a relation defined on the Euclidean plane $\mathbb{R}^2$ as follows: two points are related if and only if their distances from the origin are equal.
   (i) Show that $\sim$ is an equivalence relation (i.e., reflexivity, symmetry, and transitivity).
   (ii) Describe the equivalence class $[P]$ for any point $P \in \mathbb{R}^2$.
   (iii) What is the geometric interpretation of the partition of $\mathbb{R}^2$ induced by this relation?

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6. Let X be a non-empty set. Suppose that S is the collection of the following two types of open rays: $(a,\ \infty) = \{x \in X \mid a < x\}$, and $(\infty,\ b) = \{x \in X \mid x < b\}$. Prove that S is a subbasis, and it generates the order topology on X.

7. Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of non-empty countable sets.
   (1) Prove that the finite Cartesian product $\prod_{i=1}^{k} A_i = A_1 \times A_2 \times \cdots \times A_k$ is countable for any $k \in \mathbb{N}$.
   (2) Determine whether the infinite Cartesian product $\prod_{n=1}^{\infty} A_n$ is necessarily countable. Justify your answer.

8. Let X be a linearly ordered set. Prove that a singleton set $\{x\}$ is open in the order topology on X if and only if x has an immediate predecessor and an immediate successor, or if x is the minimum/maximum and has an immediate successor/predecessor respectively.

9. Prove that if V is closed in Y and Y is closed in X, then V is closed in X.

10. Let Y be a subspace of X and let A be a subset of Y. Prove that the topology A inherits as a subspace of Y is identical to the topology it inherits directly as a subspace of X.

11. Let A be a subset of a topological space X and $x \in X$ be an element. Then prove that the following statements are equivalent:
    (i). The point $x \in \bar{A}$.
    (ii). Every neighborhood of x intersects A.
    (iii) Every basis element containing x intersects A.

12. Prove that $\bar{A} = A \cup A'$. Deduce that A is closed in X if and only if $A' \subseteq A$.

13. Prove that Int $A \cap$ Bd $A = \emptyset$, and $\bar{A} =$ Int $A \cup$ Bd A.

14. Prove that every order topology is Hausdorff.

15. Let X be a T_1 space. Then prove that for each pair x_1, x_2 of distinct points, there exists open sets U_1 and U_2 containing x_1, and x_2 respectively such that $x_1 \notin U_2$ and $x_2 \notin U_1$.

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